



Email : aar15@rice.edu

Internship Certificate

1/09/2026

To whom it may concern,

I, the undersigned, **Ahad Rahman, Co-Founder and CTO at PULSE**, hereby certify that **Roudayna Kouka**, a software team intern at **PULSE**, completed a **summer internship program** within our company from **June 8th to September 4th, 2026**.

Internship Details:

Project Title: Design and Development of PULSE's Software Pipeline ***PulseCORE***

Mission:

During her internship, Ms. Kouka participated in the development of **PulseCORE**, building an innovative **full-stack cloud infrastructure and data pipeline solution** for PULSE's energy modeling platform. Her main tasks included:

- **Designing and provisioning cloud infrastructure with Terraform**, including BigQuery, Cloud Storage, IAM, Secret Manager, and Artifact Registry resources
- **Configuring CI/CD and deployment infrastructure**, including Cloud Run service accounts, Artifact Registry image storage, and Cloud Scheduler for secure endpoint invocation
- **Building the data storage and ingestion architecture**, including GCS Object Versioning, lifecycle retention policies, and a generic multi-source ingestion structure for the landing bucket
- **Validating and optimizing the Feature Store and caching layer**, including Redis caching, BigQuery partition pruning, and entity-key/schema integrity checks
- **Ensuring infrastructure security and disaster recovery**, including automated backup strategies, least-privilege IAM configuration, and Secret Manager integration

Ms. **Roudayna Kouka** demonstrated excellent initiative, technical rigor, and strong problem-solving skills in cloud infrastructure and DevOps engineering, as well as solid proficiency in backend and data engineering technologies. Her full-stack development skills, reliability, and dedication contributed significantly to the success of the **PulseCORE project**.

We acknowledge the quality of her work and her continuous commitment, and we wish her full success in her future studies and professional career.



A. RAHMAN
CO-FOUNDER & CTO, PULSE